

COPYBOOK

COPY Statement

Issue

Many programs may use the same file. The data names on the record might not be consistent across these programs and it makes it difficult to do code maintenance and analysis.

Solution

Create a common record layout as a copybook and use a COPY statement in a program to copy the content.

Creating a copybook

Create a PDS member in a COPYLIB, for example:

```
SLC.CPA.COMP251.COPYLIB(STUREC)
```

- ***Member name must begin with alphabetic and is max of 8 characters.***

Member STUREC contains the common record layout:

*** Student record definition

```
05  STU-NUM          PIC 99999.  
05  STU-NAME         PIC X(30) .  
05  STU-ADDRESS      PIC X(30) .  
05  STU-PHONE        PIC X(10) .
```

- Copybook is copied into a COBOL program for compilation. So, it must have valid COBOL syntax
- It must **NOT** contain another COPY statement
- It can be used for procedure division code, but mostly used for defining a record

Referencing copybook in COBOL source code

Use COPY statement as below.

In File Description

```
FD F01-INPUT-FILE.
```

```
01 F01-INPUT-REC.
```

```
    COPY STUREC.
```

Only specify the copybook member in the COPYLIB

Or in Working Storage

```
01 W01-INPUT-REC.
```

```
    COPY STUREC.
```

COBOL compiler uses SYSLIB as the DD Name for the COPYLIB. It looks for the copybooks that are used in the program being compiled.

```
//STEP020 EXEC IGYWCLG
```

```
//COBOL.SYSIN DD DSN=&SYSUID..COMP251.COBOL(LAB1),DISP=SHR
```

```
//COBOL.SYSLIB DD DSN=SLC.CPA.COPYLIB,DISP=SHR
```

Possible to use the same copybook more than once in a program

```
01  W01-INPUT-REC.  
    COPY STUREC.
```

```
01  W02-OUTPUT-REC.  
    COPY STUREC.
```

- Issue: Duplicate data names exist in the program.
- Need to use the “OF” modifier to specify the record that the names belong to.

For example:

```
MOVE STU-NAME OF W01-INPUT-REC TO STU-NAME OF W02-OUTPUT-REC
```

COPY REPLACING Clause

REPLACING performs text substitution during compilation of **COPY** statements.

SLC.CPA.COMP251.COPYLIB(STUREC)

```
05 :WNN:-STU-NUM          PIC 99999.  
05 :WNN:-STU-NAME         PIC X(30).  
05 :WNN:-STU-ADDRESS      PIC X(30).  
05 :WNN:-STU-PHONE        PIC X(10).
```

Notice the “:WNN:” tag at beginning of every name.

Write COPY REPLACING statement in COBOL source program.

```
01 W01-INPUT-REC.  
    COPY STUREC REPLACING ==:WNN:== BY ==W01==.  
01 W02-OUTPUT-REC.  
    COPY STUREC REPLACING ==:WNN:== BY ==W02==.
```

Compiler Listing for COPY REPLACING

Any lines of code that are included as a result of a COPY are identified with a “C” beside the line number in the compile listing

```
000016          WORKING-STORAGE SECTION.
000017      01  W01-INPUT-REC.
000018          COPY STUREC REPLACING ==:WNN:== BY ==W01==.
000019C      05  W01-STU-NUM          PIC 99999.
000020C      05  W01-STU-NAME        PIC X(30).
000021C      05  W01-STU-ADDRESS    PIC X(30).
000022C      05  W01-STU-PHONE     PIC X(10).
000023      01  W02-OUTPUT-REC.
000024          COPY STUREC REPLACING ==:WNN:== BY ==W02==.
000025C      05  W02-STU-NUM          PIC 99999.
000026C      05  W02-STU-NAME        PIC X(30).
000027C      05  W02-STU-ADDRESS    PIC X(30).
000028C      05  W02-STU-PHONE     PIC X(10).
```